

Comparative Evaluation of Microbiome Simulation Methods Using Real Dietswap Data

Course: Microbiome Data Analysis — Assignment 1

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1. Background

Recent advances in sequencing technologies have greatly improved our understanding of the human microbiome and its links to a wide range of diseases and biological traits. In particular, modern methods such as 16S rRNA sequencing enable rapid, parallel assessment of the relative abundances of all taxa across entire microbial communities. Although microbiome data are increasingly available, statistical analysis remains challenging. Microbiome datasets have unique characteristics that are difficult to model analytically, including sparsity (most taxa are absent in a given sample), overdispersion (the variance of read counts exceeds that assumed by standard parametric models), compositionality (the read counts within a sample sum to a constant), and strong correlations between taxa. As microbiome data have grown, a wide range of statistical methods has emerged to study these relationships. This rapid development has created a strong demand for straightforward, efficient, and reliable tools that can generate realistic microbiome simulations for rigorous methodological evaluation.

Traditional simulation approaches for microbiome data often rely on simplified assumptions that fail to reflect the complex multivariate structure observed in real datasets. Some methods capture marginal distributions but not inter-taxa correlations, while others preserve correlations but struggle with zero inflation, overdispersion, or compositionality.

MIDASim (Microbiome Data Simulator) is designed to overcome these limitations, by simultaneously modelling both the presence-absence patterns and abundance structures of a template microbiome dataset. However, studies evaluating the performance of MIDASim remain limited. The objective of this project is to systematically assess the performance of MIDASim in generating realistic microbiome data and to evaluate how closely the simulated data resemble the original template dataset. The evaluation will proceed in three distinct phases. Assessment of performance, assessment of consistency and robustness, and a comparison with other methods.

2. Methods

2.1 Template Data and Parametric Simulation (MIDASim)

We used MIDASim to generate realistic microbiome datasets by adapting an existing template dataset. The template was derived from the Dietswap study by O’Keefe et al (2015), in which 22 African American (AAM) participants were placed on a high-fibre, low-fat African-style diet and 16 rural South African (AFR) participants were placed on a high-fat, low-fibre Western-style diet under controlled conditions. For this analysis, we focused on samples collected at day 2, resulting in a dataset containing 130 taxa across 37 samples.

To generate synthetic datasets, we employed the parametric mode of MIDASim, which implements a two-step procedure: first modeling the presence-absence of taxa, and then modeling the non-zero relative abundances. In this framework, MIDASim first extracts taxon-specific characteristics from the template dataset—including mean relative abundances, marginal probabilities of zeros, dispersion estimates, and inter-taxa correlation structures—and fits a generalized gamma model to the non-zero values. The model parameters are used to preserve the structure of the original microbiome community, including the relationships between taxa and the distribution of abundances, in the simulated datasets. To control the magnitude and structure of group-level differences, we set the effect size to 1.2 and designated 10% of taxa as causal. Additionally, 70% of the taxa were randomly assigned to one subset and the remaining 30% to another, with their relative abundances adjusted in opposite directions to introduce bidirectional, group-specific shifts.

2.2 Comparative Simulation Methods

To assess the performance of the MIDASim parametric approach, we generated comparable synthetic datasets using three alternative methods: MIDASim’s nonparametric mode, SparseDOSSA2, and SPsimSeq.

A. MIDASim Nonparametric Mode

The nonparametric mode generates data directly by drawing from the empirical distribution of the template dataset. Similar to the parametric mode, the first step models the presence-absence of taxa. For taxa determined to be non-zero in this step, a rank-based approach is used to assign their non-zero relative abundances.

B. SparseDOSSA2

SparseDOSSA2 is a simulation method for generating realistic synthetic microbiome datasets by learning key characteristics from a template dataset. The method models microbial abundances using zero-inflated log-normal marginal distributions and links features via a Gaussian copula to preserve inter-feature correlation structures. Model parameters are estimated using a penalized expectation-maximization (EM) algorithm, including per-feature fraction of non-zero abundances, the mean and standard deviation of non-zero abundances, and feature-feature correlations.

C. SPsimSeq

SPsimSeq is a semi-parametric simulator originally developed for RNA-seq data but widely applied to microbiome datasets. It models count data using a negative binomial distribution to capture the high mean-variance relationship typical of sequencing data. The method separately models zero inflation to account for high sparsity and uses a gaussian copula to preserve correlations among taxa.

2.3 Assessment and Comparison

To assess the performance of MIDASim, a single simulated dataset was used. For this dataset, key taxon-level metrics—including mean abundance, variance, coefficient of variation (CV), and fraction of zeros—were calculated. These metrics capture essential microbiome characteristics: central tendency (mean), dispersion (variance and CV), and sparsity (presence-absence patterns). Relationships between these properties in the simulated and real datasets were visualized separately for the AAM and AFR groups, including:

- Mean Abundance vs. Variance
- Mean Abundance vs. CV
- Mean Abundance vs. Fraction of Zeros

To assess the consistency of MIDASim, we generated 100 independent simulated datasets. For each simulated dataset, key statistical metrics, including the mean, variance, CV, and fraction of zeros, will be calculated for each taxon. To quantify deviations, mean abundances were divided into bins, and within each bin, the difference between simulated and real datasets was computed for each metric. This process yielded 100 difference values per bin for each metric. The distributions of these differences will be visualized using boxplots or ridge plots to display the full distribution within each interval. The spread of the boxplots reflects the consistency of MIDASim simulations: smaller variance across boxplots indicates higher consistency and better reproducibility of real data characteristics across multiple simulated datasets.

Finally, the parametric version of MIDASim was compared with (a) MIDASim’s nonparametric mode, (b) SparseDOSSA2, and (c) SPsimSeq to evaluate which method best captures the key statistical properties of microbiome data, including mean, variance, CV, and fraction of zeros.

3. Results

The Dietswap dataset comprised 130 taxa (OTUs) across 37 samples, with approximately 21% zeros, indicating moderate overall sparsity. Library sizes were highly skewed across samples and the CV of the data was 5.13, suggesting substantial overdispersion. At the taxon level, CV values ranged from 0.53 to 14.90, with a median of 1.36 and a mean of 2.55.

3.1. Assessment of MIDASim Performance

Visual inspection of the relationships between mean abundance and variance and mean abundance and CV showed that the simulated data closely resembled the real data across both AAM and AFR groups (Figures 1a–1b). For the relationship between mean abundance and the fraction of zeros (sparsity), the simulation correctly reproduced the trend that low-abundance taxa were more likely to be absent. However, it deviated from the real data for taxa with intermediate abundances: in the simulations, the fraction of zeros exhibited greater variability and did not follow the distribution observed in the real dataset (Figure 1c).

To formally evaluate whether the simulated data reproduced the mean–variance relationship observed in the real dataset, we modeled the association between log-mean abundance and log-variance using a linear regression with an interaction term between mean abundance and data source (real vs. simulated). Because the mean–variance trend appeared approximately linear on the log–log scale, a linear model was appropriate:

$$\log(\text{var}) = \beta_0 + \beta_1 \times \log(\text{mean}) + \beta_2 \times \text{source} + \beta_3 \times (\log(\text{mean}) \times \text{source})$$

Where **source** indicates whether the observation comes from **Real Data** or **Simulated Data**, and β_3 represents the **difference in slopes** between the simulated and real datasets. When $\beta_3 = 0$: the simulator **perfectly matches** the slope observed in the real data, whereas values of $\beta_3 \neq 0$: the simulator **deviates** from the real-data slope.

The estimated slope difference was not significant in the AAM group, indicating that the simulated data closely reproduced the mean–variance structure of the real samples. In the AFR group, the slope difference was statistically significant but quantitatively small relative to the overall slope of the mean–variance relationship ($\beta_3 = 0.09$, SE = 0.04, p = 0.03), suggesting that although a detectable difference existed, the deviation between the real and simulated data was modest.

Table 1: Slope differences and main effects for each nationality

Nationality	Term	Estimate	Std. Error	t value	p-value
AAM	Main effects	1.93	0.04	51.38	0.000
AAM	Slope difference (Simulated vs Real)	0.00	0.05	-0.01	0.990
AFR	Main effects	1.84	0.03	56.92	0.000
AFR	Slope difference (Simulated vs Real)	0.09	0.04	2.15	0.033

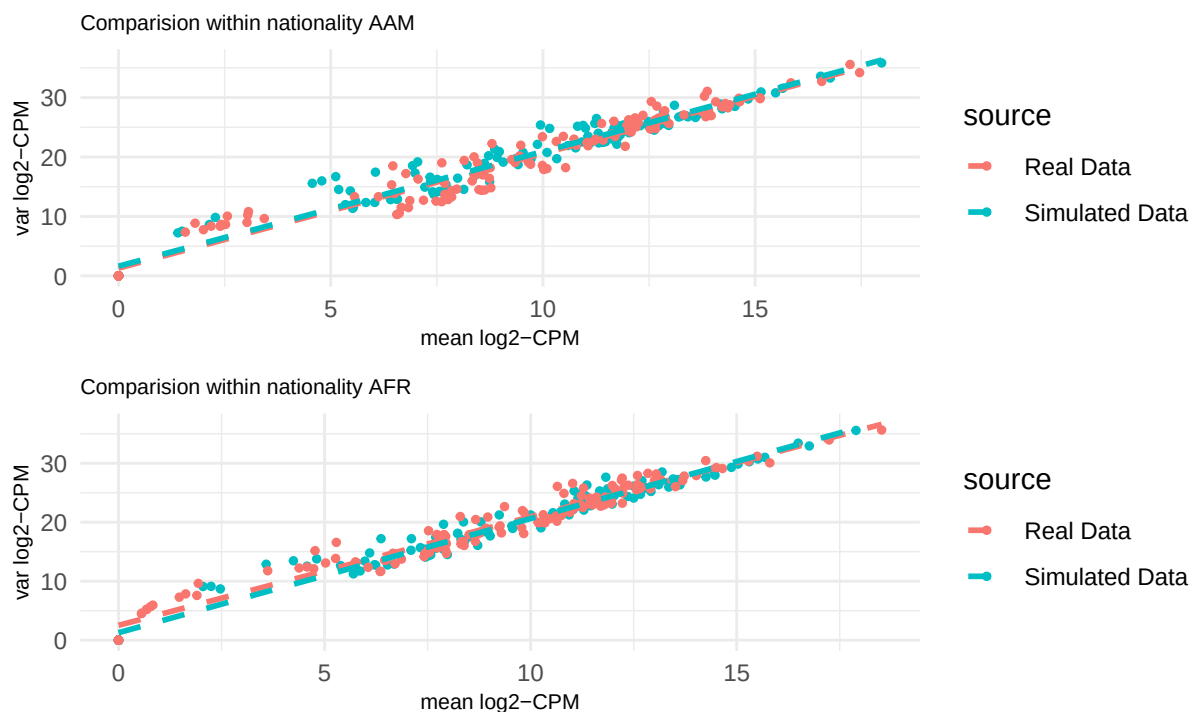


Figure 1a: Comparison of the variance versus mean relationship for taxon abundances between the real Dietswap dataset and a single simulated dataset generated using MIDASim (parametric mode). Results are shown separately for the African American (AAM) and Rural South African (AFR) groups.

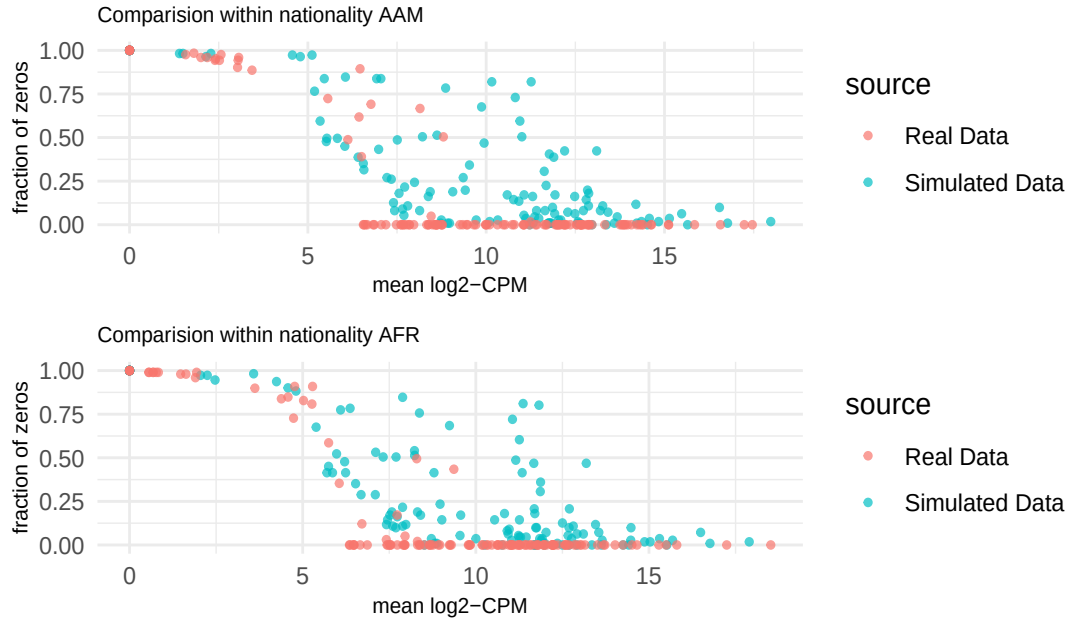


Figure 1b: Comparison of the fraction of zeros versus mean relationship for taxon abundances between the real Dietswap dataset and a single simulated dataset generated using MIDASim (parametric α). Results are shown separately for the African American (AAM) and Rural South African (AFR) groups.

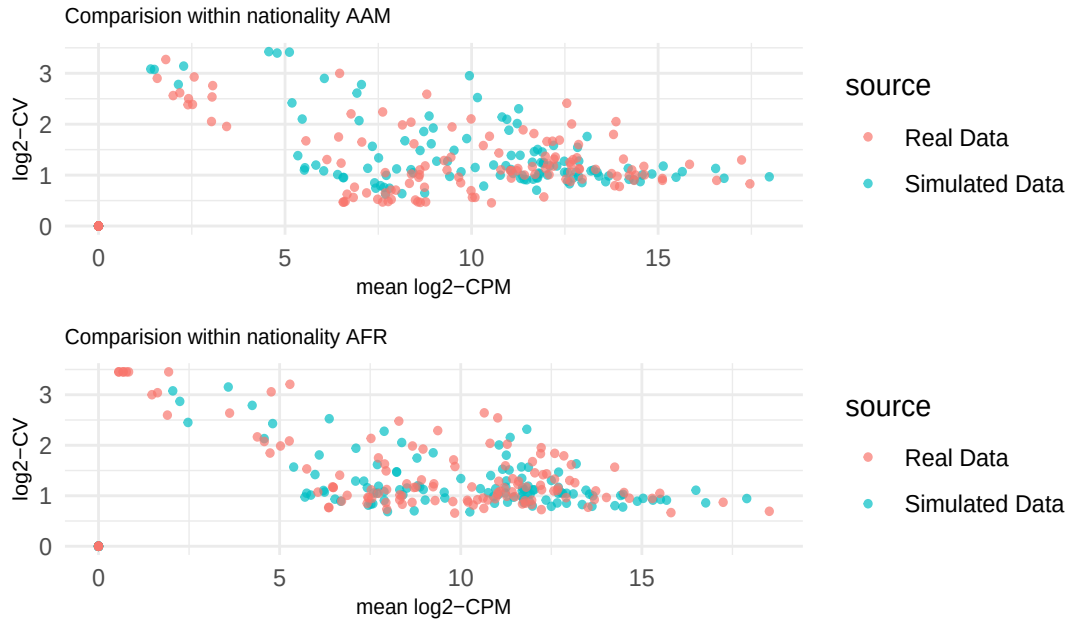


Figure 1c: Comparison of the coefficient of variation (CV) versus mean relationship for taxon abundances between the real Dietswap dataset and a single simulated dataset generated using MIDASim (parametric α). Results are shown separately for the African American (AAM) and Rural South African (AFR) groups.

3.2. Consistency Assessment

To compare the real and 100 simulated datasets across the full range of taxon abundances, we initially divided the mean abundance values into 50 narrow intervals. For each interval, we calculated the mean variance and the difference between simulated and real mean log-variance. Results for the AAM group are shown in Figure 2. The differences fluctuated widely across the 100 simulations, leading to high variability and poor reproducibility. This instability was most observed in low-abundance intervals (roughly the first 17; log-mean $\approx 0-6$) and again at very high abundances (log-mean > 15.5).

Examination of the mean-abundance distributions in both the real and simulated datasets for the AAM groups (Figure 3) showed that these narrow intervals often contained very few taxa (sometimes only 1, 2, or even none). As a result, a single simulated taxon falling into such a bin could disproportionately affect the interval’s mean variance. This scarcity of data explains the inflated variability at both ends of the abundance range.

Instead of choosing an arbitrary number of intervals (e.g., 50), we used quantile-based intervals derived from the distribution of mean abundances in the real data, ensuring that each interval contained enough taxa for stable estimates. This approach forces each interval to have roughly the same number of taxa (equal-frequency binning). We tested different minimum numbers of taxa per interval: 3 taxa per bin and 5 taxa per bin. Using 3 taxa per bin still produced unstable and noisy estimates of mean log-variance differences, even in the middle range where data are usually dense.

For 5 taxa per bin, we determined the number of intervals based on the dataset size. With 130 taxa total, approximately 26 intervals would be needed ($130/5 \approx 26$). The real data were divided into 25 quantile-based intervals, each containing roughly the same number of taxa to avoid empty or overly sparse bins. However, because many taxa had identical mean-abundance values some percentile breakpoints were identical, which “collapsed” intervals. As a result, the final number of intervals was 24. We then calculated the mean variance, CV, and fraction of zeros for the 100 simulated datasets within these 24 intervals.

Using 5 taxa per bin, variability in mean log-variance between simulated and real data decreased noticeably, resulting in more stable estimates for both nationalities (Figures 4a). However, the simulator did not fully capture the zero-inflation observed in the real data, although the fraction of zeros was reasonably well reproduced in the tails (extreme low/high abundances) (Figures 4b), consistent with patterns seen in a single simulated dataset (Figure 1b). CV was poorly reproduced but still better than the fraction of zeros (Figures 4c). In other words, the simulator reproduced variance patterns reasonably well but struggled to replicate the strong sparsity of the real data.

3.3. Comparative Performance Analysis

We compared how well different simulation methods reproduce key statistical properties of the real microbiome data, focusing on three relationships: mean abundance versus variance, mean abundance versus coefficient of variation (CV), and mean abundance versus fraction of zeros.

For the non-parametric MIDASim simulations, we applied a filtering step to remove very low-abundance taxa, which resulted in 107 taxa. In contrast, SparseDOSSA2 and SPsimSeq handled rare taxa better, retaining 114 taxa. To ensure a fair comparison across all methods, the corresponding taxa in the real dataset were filtered to match those present in the simulated datasets (Figure 5a-5c).

Overall, methods that explicitly account for non-parametric distributions or semi-parametric approaches more accurately reproduced the key statistical patterns observed in the real data, particularly the relationship between mean abundance and the fraction of zeros (Figure 5b). However, they still show some limitations in fully capturing the behavior of very rare taxa.

The parametric version of MIDASim, while able to model the general structure, was less effective at capturing variability in low-abundance taxa and zero-inflation patterns. The simulated points were highly scattered in the low- to mid-abundance range (Figure 5b, panel a).

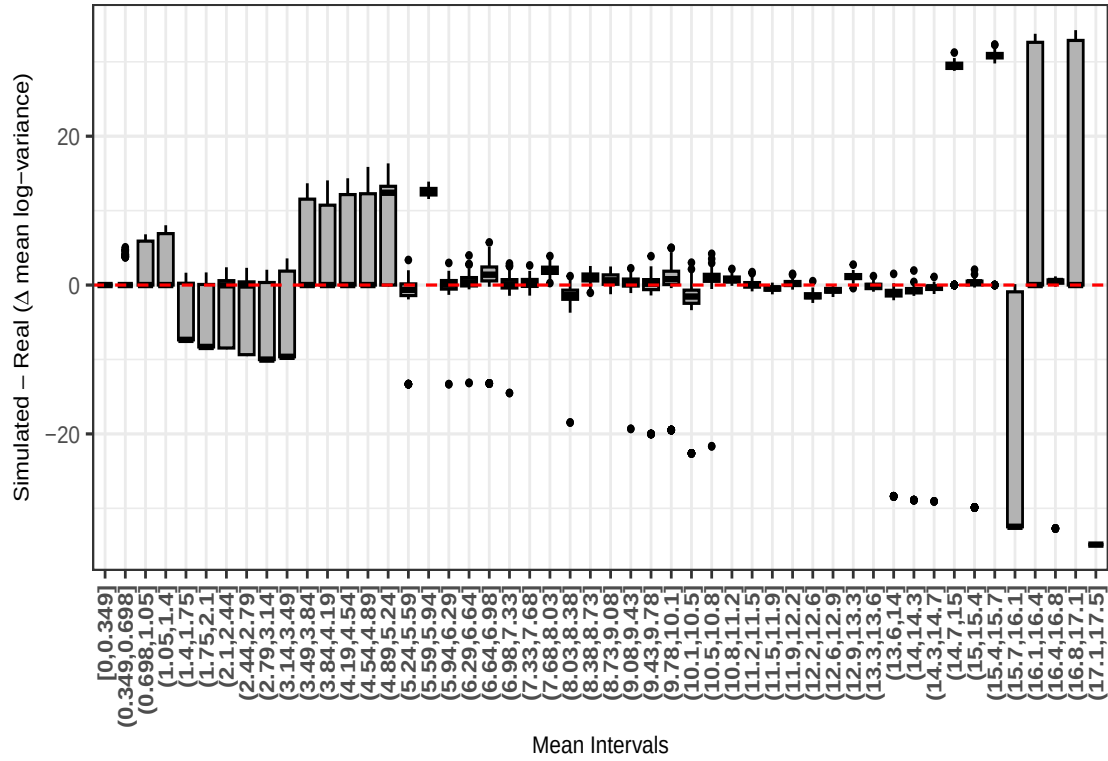


Figure 2: Distribution of the differences between simulated and real mean log-variance across 50 mean-abundance intervals, based on 100 simulations (AAM group).

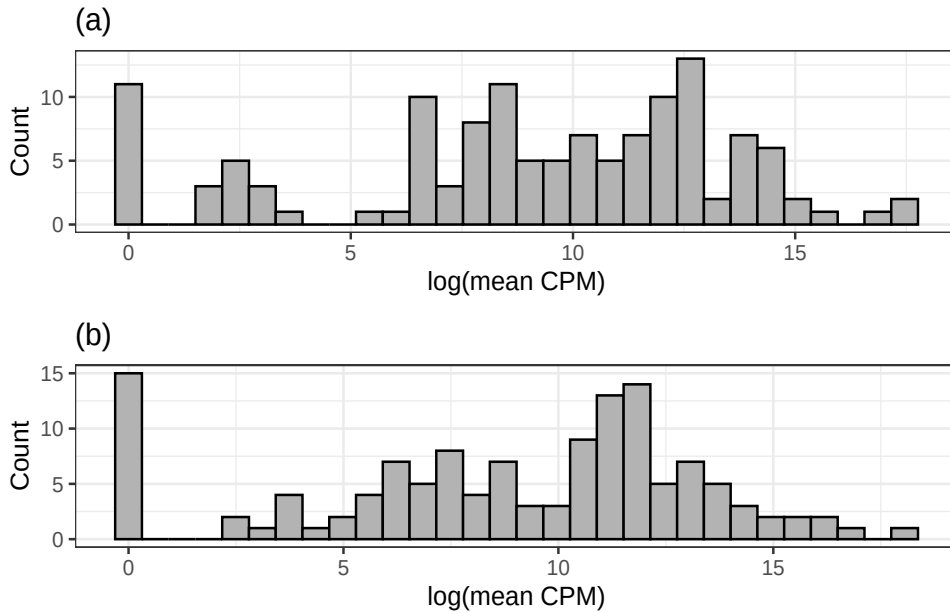


Figure 3: Distribution of log-mean-CPM (a: real data, b: single simulated data).

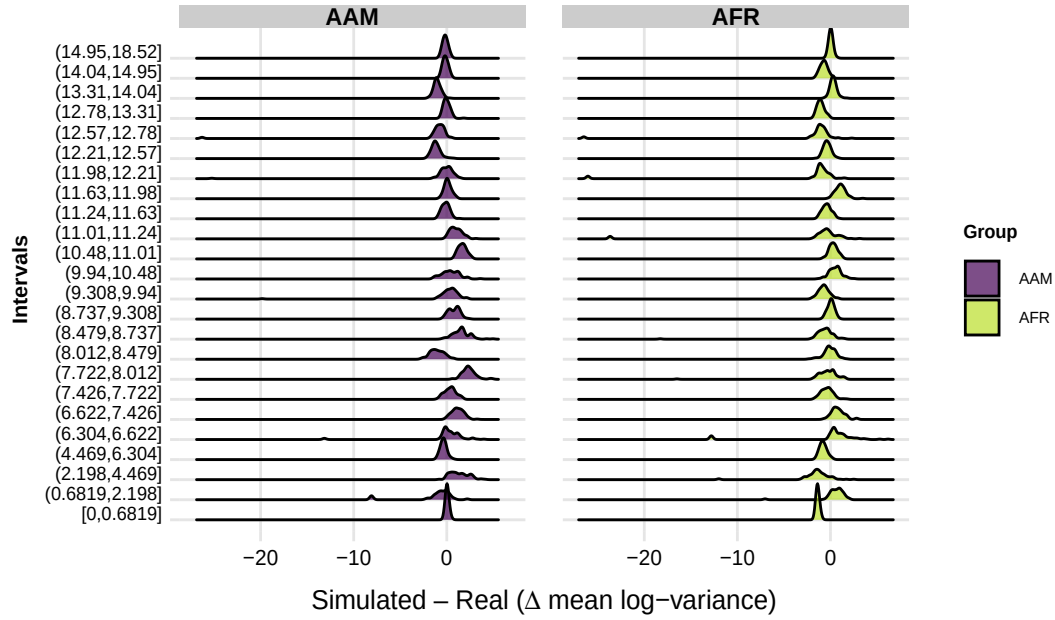


Figure 4a: Distribution of differences between simulated and real mean log-variance across 24 mean-abundance intervals (minimum 5 taxa per interval), based on 100 simulations for for both AAM and AFR.

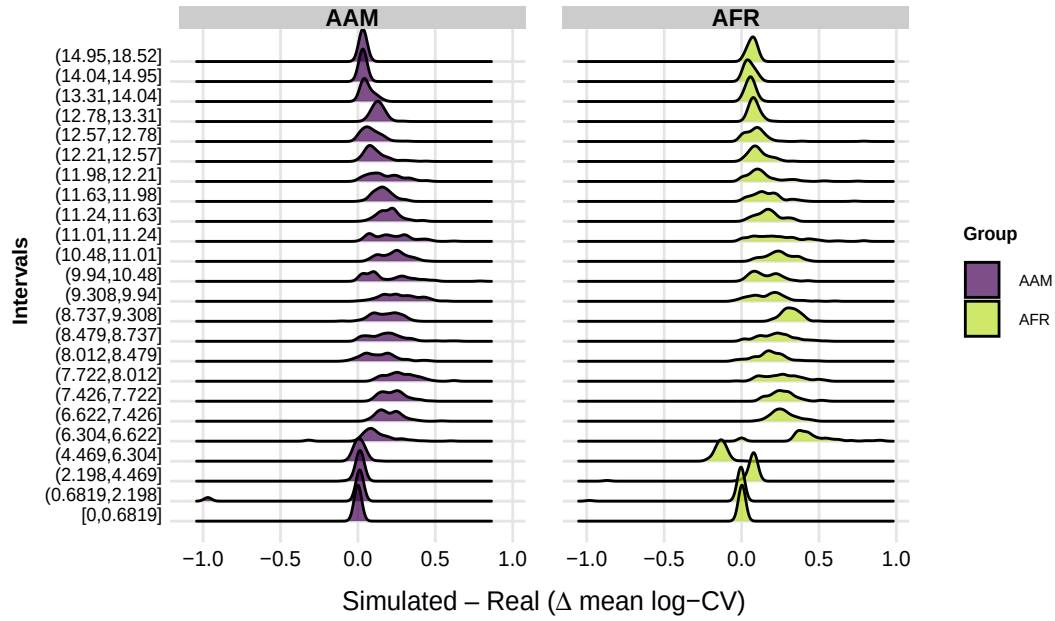


Figure 4b: Distribution of differences between simulated and real mean log-fraction of zeros across 24 mean-abundance intervals (minimum 5 taxa per interval), based on 100 simulations for for both AAM and AFR.

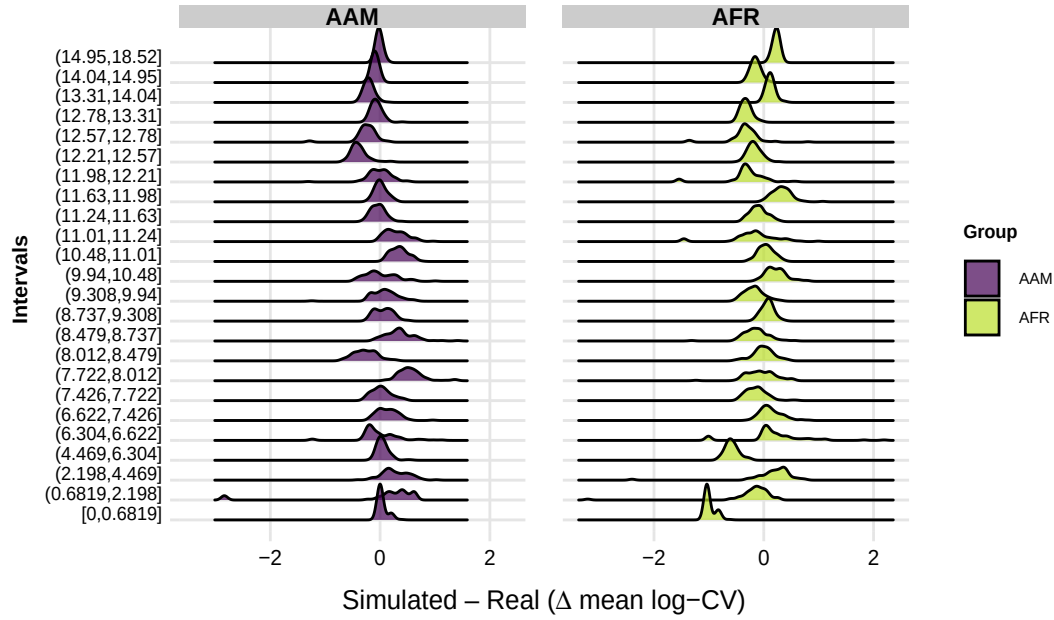


Figure 4c: Distribution of differences between simulated and real mean log-CV across 24 mean-abundance intervals (minimum 5 taxa per interval), based on 100 simulations for for both AAM and AFR.

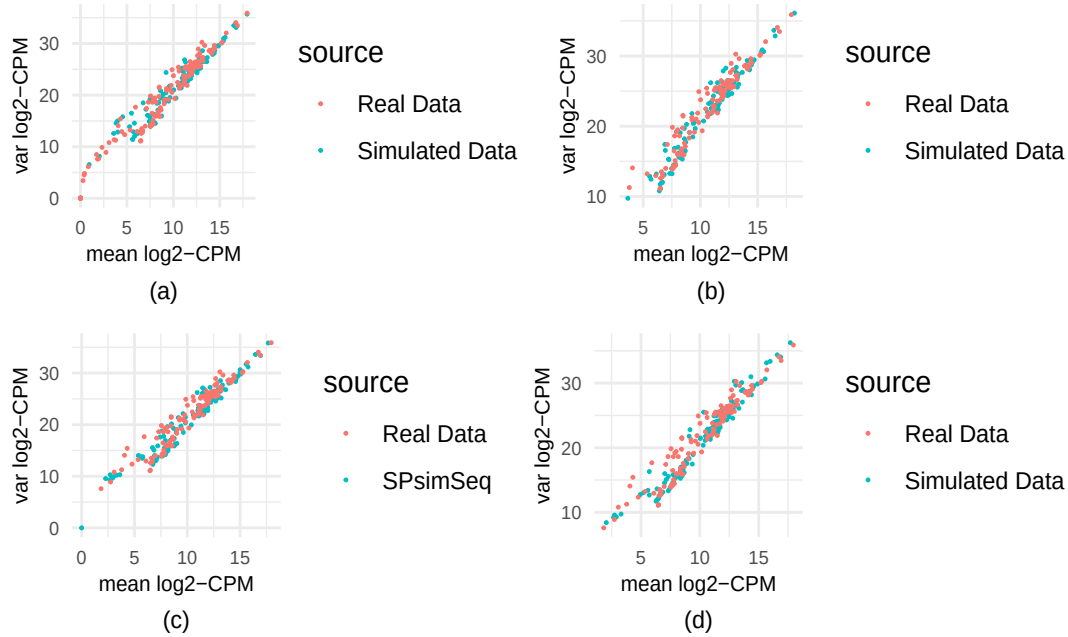


Figure 5a: Comparison of the variance-mean relationship for taxon abundances between the real dataset and single simulated datasets across different simulation methods
a: MIDASim (parametric), b: MIDASim (non-parametric), c: SPsimSeq, d: SparseDOSSA2

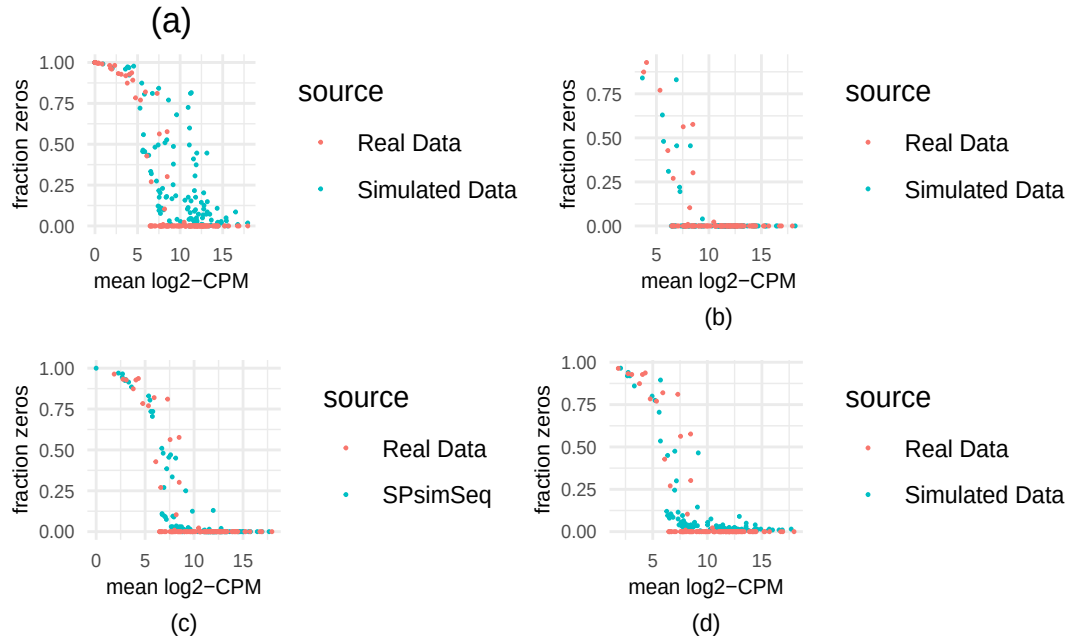


Figure 5c: Comparison of the fraction of zeros vs mean relationship for taxon abundances between the real dataset and single simulated datasets across different simulation methods
a: MIDASim (parametric), b: MIDASim (non-parametric), c: SPsimSeq, d: SparseDOSSA2

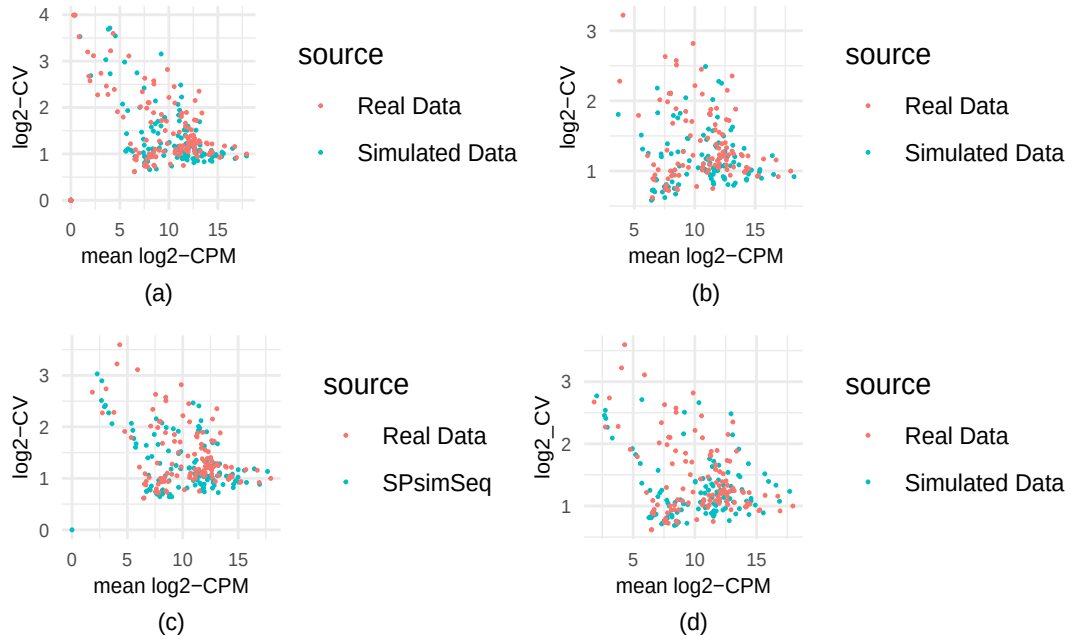


Figure 5b: Comparison of the CV-mean relationship for taxon abundances between the real dataset and single simulated datasets across different simulation methods
a: MIDASim (parametric), b: MIDASim (non-parametric), c: SPsimSeq, d: SparseDOSSA2

4. Conclusions

Both single and 100 parametric-mode simulations closely resembled the template data, particularly reproducing the mean–variance structure; however, the simulator was less accurate at capturing sparsity. Non-parametric simulations performed better in replicating the real data patterns and make relatively few distributional assumptions about relative abundances. Nonetheless, non-parametric methods require a minimum number of non-zero counts per taxon to estimate distributions reliably, which limited the number of simulated samples compared to the original dataset.

4. Reference

O’Keefe, S. J., J. V. Li, L. Lahti, J. Ou, F. Carbonero, K. Mohammed, J. M. Posma, J. Kinross, E. Wahl and E. Ruder (2015). “Fat, fibre and cancer risk in African Americans and rural Africans.” *Nature communications* 6(1): 6342.

Assefa, A. T., J. Vandesompele and O. Thas (2020). “SPsimSeq: semi-parametric simulation of bulk and single-cell RNA-sequencing data.” *Bioinformatics* 36(10): 3276-3278.

He, M., N. Zhao and G. A. Satten (2024). “MIDASim: a fast and simple simulator for realistic microbiome data.” *Microbiome* 12(1): 135.

Assefa (2022) provided the SPsimSeq code on GitHub (<https://github.com/CenterForStatistics-UGent/SPsimSeq/blob/master/R/SPsimSeq.R>, accessed 2025-12-05).

Ma (2021) provided the SparseDOSSA2 code on GitHub (https://github.com/biobakery/SparseDOSSA2/blob/master/R/SparseDOSSA2_fit.R, accessed 2025-12-05).

He (2025) provided the MIDASim code on GitHub (<https://github.com/mengyu-he/MIDASim/tree/master/R>, accessed 2025-12-05).